

Siddharth Magesh

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CAREER OBJECTIVE

To pursue advanced research in Artificial Intelligence, with a focus on Natural Language Processing, Computer Vision, and Generative AI systems. I aim to contribute to cutting-edge research in Large Language Models and multimodal AI architectures that bridge theoretical foundations with practical implementations. My ultimate goal is to advance the field toward Artificial General Intelligence through innovative research and development.

EDUCATION

- Expected 2026 B.Tech in Artificial Intelligence and Data Science**
Velammal Engineering College, Anna University, Chennai, India
CGPA: 8.91/10.0
- Expected 2026 Minor in Full Stack Development**
Velammal Engineering College, Anna University
CGPA: 8.91/10.0
- 2022 Higher Secondary Certificate (12th Standard)**
Velammal Vidhyashram, CBSE Board
Percentage: 85%
- 2020 Secondary School Certificate (10th Standard)**
Narayana E-Techno Kolathur, CBSE/IIT
Percentage: 82%

RESEARCH EXPERIENCE & PROJECTS

- AI Research Intern – Scrapify Ecotech (Ecofloater Project)** 2024-2025
Intern Research - Completed
Developed a fully autonomous Ecofloater system for river cleaning using custom computer vision models and synthetic datasets. Integrated stereovision and depth mapping with Intel RealSense cameras, applied reinforcement learning for adaptive decision-making, and deployed the optimized pipeline on NVIDIA Jetson Nano for real-time inference.
- OSSARTH – AI-Powered Operating System** 2024-2026
Research Paper - Ongoing Project
Designed a natural language-driven operating system replacing traditional GUI/CLI. Built a lightweight, optimized system for low-power devices (e.g., Raspberry Pi) with real-time file management, program execution, and modular plugin support, enabling fast interaction without complex command memorization.
- MTERBS – Intelligent Emergency Traffic Control System** 2023-2026
Research Paper - Ongoing Project
Developed an AI-integrated traffic control framework using deep learning and IoT-enabled vision sensors for real-time traffic monitoring and intelligent ambulance prioritization. Implemented adaptive lane clearance, timed signal release, and dynamic one-way allocations to minimize golden-hour delays while ensuring smooth traffic flow and maintaining post-incident stability.

TECHNICAL SKILLS

Programming Languages	Python, C/C++, Java, Assembly (x86)
AI/ML Research	Deep Learning Architectures, Neural Network Optimization, Model Fine-tuning, Transfer Learning, Few-shot Learning, Attention Mechanisms, Transformer Architectures
AI Specializations	Large Language Models, Computer Vision, Natural Language Processing, Generative AI, Model Context Protocol, Agentic AI Models, Retrieval-Augmented Generation (RAG)
ML Frameworks	PyTorch, TensorFlow, Hugging Face Transformers, LangChain, OpenAI API, Anthropic Claude API
Backend Engineering	REST APIs (Flask, FastAPI), Node.js, Express.js, Microservices Architecture, API Gateway Design, Postman, Podman
Databases	MySQL, MongoDB, MongoDB Atlas, CRUD Operations, Database Design, Query Optimization
MLOps & DevOps	Git, AWS SageMaker, Docker, UV, Linux, CI/CD Pipelines, Model Deployment, Version Control

PROFESSIONAL EXPERIENCE

AI/ML Intern – Inspire Solutions (Aladipattiyar Karupatti Coffee)	2023-2024
Built machine learning models for analyzing stock handling and verifying original supply amounts for restaurant operations. Developed computer vision models for automated cleanliness assessment of utensils and sanitization monitoring throughout the establishment. Created vision-based systems for real-time shelf inventory monitoring and implemented various automation tasks for restaurant management. Contributed to multiple open-source datasets for hospitality industry AI applications.	
Lead Backend Developer – Velammal Engineering College Website	2024-2025 (Ongoing)
Led a team of backend developers in building comprehensive college website infrastructure and designed system architecture for scalable web applications. Managed collection and processing of large-scale institutional data while implementing robust security measures and performance optimization features. Developed code modularization strategies and created automation scripts with CI/CD pipelines for streamlined development workflows.	
Co-Founder – GradGear (AI-Powered Consultancy)	2024-2025 (Ongoing)
Co-founded startup developing AI-driven application for personalized laptop recommendations based on user requirements, budgets, and use cases. Built dynamic web scraping tools and established local vendor network connections for real-time pricing data aggregation. Developed comprehensive analysis algorithms that generate detailed reports and tailored recommendations for optimal technology purchasing decisions.	

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

• Introduction to Generative AI (Duke University)	Apr 23, 2025
• Foundational Level in Programming and Data Science (IITM)	Apr 30, 2025
• Assembly Language Programming (Udemy)	Jul 23, 2025
• TensorFlow for AI, ML, and Deep Learning (Coursera)	Nov 17, 2024
• Generative AI with Large Language Models (Coursera)	Aug 15, 2024
• Natural Language Processing - 4 Course Specialization (Coursera)	Jul 27, 2024

- **Python for Data Science** (NPTEL) Feb 01, 2024
- **Supervised Learning: Regression and Classification** (Coursera) Dec 08, 2023
- **Advanced Diploma in Java Programming** (Computer Software College) Nov 05, 2022
- **Power BI and Tableau** (Imarticus Learning) Apr 01, 2024

PROJECTS

Sentimatrix – Multi-modal Sentiment Analysis Platform **2023-2024**

Major Project — Published

AI-powered sentiment analysis and product intelligence tool designed to help businesses understand public sentiment across multiple channels. Combines multimodal sentiment analysis, autonomous web scraping, and interactive reporting capabilities. Developed comprehensive ecosystem including PyPI library, RESTful API, intelligent chatbot interface, and Business Analytics Tool (BAT) website for sentiment monitoring and decision-making support.

AgriHub – Integrated Agricultural Platform **2022-2023**

Major Project — Deployed

Comprehensive digital platform addressing multiple agricultural challenges through a unified solution. Integrates retail management, research databases, educational resources, and peer-to-peer guidance systems for farmers. Implemented AI-driven crop recommendation algorithms, IoT sensor integration, machine learning models for yield prediction, and blockchain-based supply chain transparency to enhance agricultural productivity.

Breathy-Sweet – IoT Health Monitoring **2023-2024**

Minor Project — Patent Pending

Non-invasive glucose monitoring system leveraging IoT sensors and advanced machine learning algorithms for diabetic care management. Utilizes breath analysis technology combined with environmental sensors to predict glucose levels without blood samples. Developed real-time data processing pipelines, predictive analytics models, and Progressive Web Application interface for continuous health insights and personalized diabetic management.

MediNote-AI – AI Medical Assistant **2024-2025**

Minor Project — In Deployment Stage

Intelligent healthcare automation system that streamlines clinical documentation through real-time conversation monitoring. Utilizes natural language processing and speech recognition to analyze doctor-patient interactions, automatically generating accurate prescriptions and medical records. Implemented transformer-based models for medical entity extraction and secure data processing pipelines to enhance clinical workflow efficiency.

ACHIEVEMENTS & RECOGNITION

Hackathons & Competitions

- **Title Winner** – INNOTHAN'24, KCG College of Technology Sep 21, 2024
- **Special Recognition Award** – PRABHIGNYAN, Prathyusha Engineering College Mar 22, 2024
- **Third Prize** – Tech Genesis'25, Velammal Engineering College Mar 17, 2025
- **Second Prize** – Ideathon, Ananta Science Society Sep 21, 2023
- Active participant in Intel GenAI Hackathon'24, HackVerse'24 (SRM), and multiple national-level technical competitions

Academic Presentations & Publications

- Research presentation at NCAIBIA'25 National Level Conference on AI Applications Mar 10, 2025
- Technical paper presentation at Hackerz Conference, Chennai Institute of Technology Feb 03, 2024

LEADERSHIP & SERVICE

Technical Head – Youth Red Cross, Velammal Engineering College 2022-2025

- Lead technical initiatives and digital transformation projects for the college YRC unit
- Previously served as Joint Secretary (2023-24) and Volunteer (2022-23)
- Developed mobile application for YRC and multiple software solutions, including LifeConnect blood donor platform, automated event management system, and official YRC website
- Authored comprehensive operational protocols and a rule book for organizational procedures and digital integration

Vice Chairman – WebOps Club, Velammal Engineering College 2024-2026

- Oversee web development projects and technical workshops for the student community
- Mentor junior students in full-stack development and deployment practices

Blockchain Club – Member, Velammal Engineering College 2022-2023

- Participated in blockchain technology workshops and decentralized application development
- Collaborated on cryptocurrency and smart contract projects with peer learning initiatives

Coders Club – Member, Velammal Engineering College 2022-2026

- Active participant in competitive programming contests and algorithm challenges
- Contributed to open-source projects and collaborative coding initiatives within the college community

Community Service

- Regular blood donation volunteer and cancer awareness campaign participant
- Environmental conservation initiatives, including lake cleaning and plantation drives
- Road safety awareness campaigns and fire safety education programs